Metals

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain ferrous, non-ferrous and alloy
- ☐ describe the properties and uses of iron, cast iron, mild steel and stainless steel
- ☐ describe the properties and uses of high-carbon steel
- ☐ describe the properties and uses of aluminium, copper and brass

I DO — Worked example

Read how a full-mark answer is built

A camping kettle has an aluminium body and a stainless steel handle.

1 Explain two reasons why aluminium is suitable for the kettle body. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Lightweight, so the kettle is easy to carry when camping
- ✓ Good conductor of heat, so the water boils quickly

Why marks are not awarded

- ✗ "Strong" or "light" with no link to the kettle

WE DO — Guided practice

Use the starters and word bank to help you

2 Explain the difference between a ferrous and a non-ferrous metal. Give an example of each. (2 marks)

Sentence starters

One reason is … because …
Another reason is … so …

Word bank

Copper · ferrous · Brass · contain · Aluminium · Mild steel · iron · metals

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Why is aluminium used for drinks cans?
☐ It is magnetic, so cans are easy to sort ☐ It is hard and stiff, so cans keep their shape ☐ It is the cheapest metal available ☐ Lightweight and corrosion resistant

b) Which metal is ferrous?
☐ Mild steel ☐ Brass ☐ Aluminium ☐ Copper

c) Brass is an alloy of copper and...
☐ Tin ☐ Zinc ☐ Iron ☐ Aluminium

d) Cast iron is unsuitable for a product that must bend because it is...
☐ Brittle ☐ Ductile ☐ Malleable ☐ Lightweight

4 Key terms: write the letter of the matching description next to each term.

1. **Ferrous metal**	____	**A** A metal mixed with other elements to improve its properties.
2. **Alloy**	____	**B** Lightweight with good strength-to-weight ratio. Cans, ladders, aircraft.
3. **Mild steel**	____	**C** Contains iron; magnetic (except stainless); rusts.
4. **Stainless steel**	____	**D** Ductile, malleable, easily welded. Car bodies, nuts and bolts.
5. **Aluminium**	____	**E** Corrosion resistant and hygienic. Cutlery, sinks.

5 Give the properties and one use of Brass. (2 marks)

6 Give the properties and one use of Copper. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?